

The G Hardware Design Language

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G hardware design language (GHDL), where G has no meaning, might remind us of very high-speed integrated circuit hardware description language (VHDL). Could these be twins? Jokes apart, VHDL is a programming language, while GHDL is a compiler to that language. This means that codes that are generated by VHDL can be tested with the help of GHDL. Let us find out more about this tool to identify its working and benefits for our projects.

GHDL is an open source tool compatible with Windows, Linux and Mac OS X operating systems (OSes). It is a free software that can be redistributed or modified under GNU GPL version 2 or any later version. The latest version of GHDL is 0.31, which is available as mcode, whereas the binary version of GHDL is still available in version 0.29.

Compiler or simulator

A simulator usually requires other tools for compilation, whereas GHDL is mainly a compiler that can execute almost any VHDL program. It converts VHDL codes directly into the machine language using the GNU compiler collection (GCC) framework. It does not require any intermediary language like C or C++ for the same. Thus, codes are compiled faster and analysed in a shorter time as compared to the compiler that uses intermediary languages. However, the mcode version of GHDL has its own JIT compiler framework.

Some popular simulators have the code synthesis ability that helps these create netlists. However, GHDL does not have the ability to create any netlist. So, rather than calling it

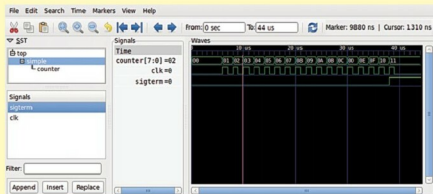


Fig. 1: Using GHDL for interactive simulation

What is mcode version?

Ideally speaking, we can divide GHDL installation package into two different installation versions—mcode and GCC. The binary version and GHDL installers are based on GCC framework and so can be called GCC versions. The mcode version of the GHDL installation package is based on JIT compiler framework. mcode means machine codes that allow a quick compilation and are easy to install on systems where GCC installation is difficult. It is limited to 32-bit Intel x86 architecture processors, and is a bit slow as compared to the GCC version.

Competitive tools

- activeHDL
- ModelSim XE
- Tarang EDA
- VHDL Simili
- Xilinx ISE

a simulator, we may call it a compiler that can also simulate.

Features to strive for

GHDL has gained popularity because of its features. It is not just limited to one processor, but can successfully compile and simulate LEON1 SPARC processor and DLX processor.

Self inspects files. GHDL can directly create binaries or executable images that are considered to be the best form for test benches because these are autonomous, self-checking VHDL designs. Binary versions can also be used to create either a VCD file or a GHW file that can be inspected visually with a waveform viewer.

Accepts codes in any language. GHDL, being a compiler, can gener-

Pre-requisites for installation

For installation of the binary version:

- C compiler
- zlib

For installation of the mcode version:

- gtkwave waveform viewer

ate an object file for codes written in any foreign language, such as C, C++ or Ada95. All versions of GHDL above 0.5 have this capability.

IVI supports GHDL. IVI is a graphical front-end for various simulators. GHDL, a compiler without any provision to view graphics, needs a separate graphical interface. Currently, versions of this tool have been enhanced to work with IVI.

Portable. GHDL can easily be ported to other OSes or CPUs without any issue; an example is porting of this tool to MAC. You can find the MAC version of the tool available for download, although the installation